N21 Channel coding

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1

Entropy of a probability distribution $p(x)$ is defined as

$$S = -\sum_x p(x)\log(x) \tag{1.1}$$

The definition of relative entropy/Kullback-Lieber divergence between two probability distributions $\{p(x)\}$ and $\{q(x)\}$ is given by

$$D\left(p(x) \parallel q(x)\right) = \sum_x p(x)\log\frac{p(x)}{q(x)} \tag{1.2}$$

For random variables X, Y, with joint probability $p(x,y)$ and marginal distribution $p(x) = \sum_y p(x,y)$ and $p(y) = \sum_x p(x,y)$, the **mutual information** is defined as

$$I(X;Y) = D\left((p(x,y) \parallel p(x)p(y)\right) \tag{1.3}$$

since $p(x,y) = p(y,x)$ and $p(x)p(y) = p(y)p(x)$ and relative entropy is non-negative, we have the following properties for the mutual information

$$I(X;Y) = I(Y;X), \qquad I(X;Y) \geq 0 \tag{1.4}$$

$I(X;Y) = 0$ when X and Y are independent variables $p(x,y) = p(x)p(y)$. More explicitly, the mutual information canbe written as

$$I(X;Y) = \sum_{xy} p(x,y)\log\frac{p(x,y)}{p(x)p(y)} \tag{1.5}$$

with a little bit of algebra one can show

$$I(X;Y) = S(X) + S(Y) - S(X,Y) \tag{1.6}$$

where $S(X)$ is the entropy of $p(x)$ and similarly for $S(Y)$, $S(X,Y)$ is the entropy of the joint distribution $p(x,y)$

$$S(X,Y) = -\sum_{xy} p(x,y)\log p(x,y) \tag{1.7}$$

the conditional entropy of X given $Y = y$ is

$$S(X|y) = -\sum_x p(x|y)\log p(x|y) \tag{1.8}$$

and the conditional entropy of X given Y is the above averaged over y

$$\begin{aligned}
S(X|Y) &= -\sum_y p(y) \sum_x p(x|y) \log p(x|y) \\
&= -\sum_{xy} p(x,y) \log p(x|y) \\
&= -\sum_{xy} p(x,y) \log \frac{p(x,y)}{p(y)} \\
&= -\sum_{xy} p(x,y) \log p(x,y) + \sum_{xy} p(x,y) \log p(y) \\
&= S(X,Y) + \sum_y \left(\sum_x p(x,y) \right) \log p(y) \\
&= S(X,Y) + \sum_y p(y) \log p(y) \\
&= S(X,Y) - S(Y)
\end{aligned} \tag{1.9}$$

where we used $p(x,y) = p(x|y)p(y)$. Rearanging yields

$$S(X,Y) = S(X|Y) + S(Y) \tag{1.10}$$

similarly

$$S(X,Y) = S(Y|X) + S(X) \tag{1.11}$$

which says we can learn the total (average) information, by learning X, then leaning Y given that we have learnt X, or equivalently first leaning Y then learning X given that we have learnt X, this makes intuitive sense. The mutual information expression Eq.(1.6) can now be rewritten as

$$I(X;Y) = S(X) - S(X|Y) = S(Y) - S(Y|X) \tag{1.12}$$

so from $I(X;Y) \geq 0$ we have

$$S(X) \geq S(X|Y), \qquad S(Y) \geq S(Y|X) \tag{1.13}$$

which says average information of X cannot be less than the average information of X after you have learnt Y, and similarly average information of Y cannot be less than Y after you have learnt X and two are the same. Mutual information is a measure of dependance of X and Y and also how much of our ignorance is reduced regarding one variable after we have learnt the other one.

2

A channel takes in an input x with that is drawn from some input distribution $p(x)$. Then it defines a conditional probability distribution $p(y|x)$, and finally output y with probability $p(x,y) = p(y|x)p(x)$. For example, a binary symmetric channel takes in a bit which is either 0 or 1, then have a 0.1 probability of flipping the bit, i.e. $p(y = 1|x = 0) = p(y = 0|x = 1) = 0.1$. $f = 0.1$ is the error probability of the channel. Ideally, we want to transfer a bit across channel/communicate with negligibly small error. A simple

scheme is to repeat the input N times (e.g. 10 times, so 1111111111), then if the output has more than half of one type of bit (e.g.1011011111), then we infer the input is the majority, this is substantially more reliable because the probability of flipping 5 or more bits is very small on the order of 0.1^5, however, this is also N times slower, if we were able to send 1 bit per second before, then now we are sending 1 bit every 10 seconds (the other 9 seconds are sending redundant bits for checking the error). The intuition is that in order to achieve infinitely small error, we need to communicate through the channel at an infinitely slow rate, remarkably, Claude Shannon was able to show we can achieve essentially error free communication at a finite rate. We shall find out what this rate is. Define the **capacity** of a channel as

$$C = \max_{\mathcal{P}_X} I(X;Y) \tag{1.14}$$

which means the maximum mutual information for the set of all possible input distributions $\mathcal{P}_X = \{P(X) = \{p(x)\}\}$.